



Year	Topic No.	Topic Name	Lesson No.	Lesson Name
4	1	Variation and classification	1	Shared features
Curriculum objectives	B4.1A – Explain how living things can be classified according to shared features LSE3.2 – Investigating: Make systematic and careful observations			
Learning focus	Groups of living things share features			
Key vocabulary	scientists; groups; similar; features; share			
Book references	Textbook pages 4–5		Workbook pages 4–5	
Resources supplied	PIPS Y4 TP1 RS1: Shared features: Invertebrates			
Other resources needed	Per learner: scissors A5 white paper – for drawing and writing pencils			

Lesson Outline

Before you teach:

In previous years, learners have grouped living things by observing their shared features. In this topic, you will develop their understanding of this by formalising it as 'classification'. Note that the term 'classify' is not introduced in the vocabulary until the next lesson, but the observation of shared features is addressed in this lesson in preparation for this.

In advance of the lesson, you will need to print off the resource sheet 'Shared features: Invertebrates', enough for one per learner. Note that these animals are not intended to be biologically accurate representations of a particular invertebrate or invertebrate group.

Introduction:

Remind learners how they have **grouped** living things before, for example, animals/plants or vertebrates/invertebrates. Also, recap that they know how **scientists** group living things according to **features** they **share**, for example, animals that have feathers are grouped together as birds. Remind them that plants that have flowers are grouped together as 'flowering plants'. However, in Year 1, they looked at how flowering plants were then put into smaller groups called 'flower families'.

Main activity:

Look at textbook pages 4 and 5 together and ask the questions about the three sets of living things circled there. Omit the bird and butterfly section for now and revisit this in the plenary. Ensure that learners understand the difference between the words '**similar**' and 'same'. An example might be to describe the class as being of similar heights but not the same height.

Give each learner a copy of the resource sheet 'Shared features: Invertebrates', a pair of scissors, and a pencil and paper. Before any cutting is done, look at the intact sheet together, explaining that they are imaginary animals not real ones. Set a scene where learners are pretending to have just discovered them and are helping scientists to group them. To ensure that everyone understands the sort of detail they should observe and compare, ask:

- *Do you think these animals are more similar to vertebrates or invertebrates?* (Invertebrates.)
- *How many legs do they have?* (Six.) Reason that this is a shared feature – they all have the same number of legs and so that feature would put them all in the



same group.

Explain that learners are looking for features that some of the invertebrates share but some do not – those features would put them into *different* groups.

Direct learners to cut round all the invertebrates – not in detail, but just to separate them. Ask the learners to:

- sort them into big groups first;
- look for any ways to group them within these bigger groups;
- write some features that each of their groups share and put these with the pictures of that group; and
- draw another new invertebrate that they would need to put into a new big group of its own.

Give everyone time to get to this point. Those who finish quickly could then draw one extra animal to go into each of their four big groups. These must be similar enough to fit the group but not an exact copy of something already there from the sheet.

When everyone has finished the basic task, group learners in pairs and ask them to compare how they each grouped the animals.

Workbook:

Complete workbook pages 4 and 5.

Plenary:

As a whole group, look at textbook page 5 together and discuss why the bird and the butterfly are not in the same animal group. Start by looking at the features of each. Use this as an opportunity to point out that living things that are grouped together share features. Here, the butterfly and the bird are both animals, but the similarity stops there. Although they both have wings, these differ a lot, and their other features are much too different to put them in a smaller group together.

Support	Learners who would not finish the grouping and drawing tasks by writing the features of their group could do this part orally.
Extension	Make an identification key for your groups. Do this at the stage where there are at least four groups.
Homework / Follow-up	Finish any incomplete workbook tasks or the identification key (see 'Extension' activity).